Renewable Energy 232 (2024) 121020

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Modeling sorption-enhanced methane synthesis for system control and operation

Andrea Barbaresi [a,*], Mirko Morini [a,b], Agostino Gambarotta [a,b], Florian Kiefer [c], Panayotis Dimopoulos Eggenschwiler [c]

[a] Department of Engineering for Industrial Systems and Technologies (DISTI), University of Parma, Parco Area delle Scienze 181/A, 43124, Parma, Italy
[b] Center for Energy and Environment (CIDEA), University of Parma, Parco Area delle Scienze 181/A, 43124, Parma, Italy
[c] Empa, Swiss Federal Laboratories for Materials Science and Technology, Ueberlandstrasse 129, Duebendorf, 8600, Switzerland

ARTICLE INFO

Keywords:
Modeling
Methanation
Energy carrier synthesis
Sorption-enhanced processes

ABSTRACT

To address energy transition in the immediate future, synthetic energy carriers are among the most promising options for decarbonization due to their capability to store renewable energy in the long term and rely on existing infrastructure. Methane, as a substitute for natural gas, can be produced at high purity through sorption-enhanced processes.

These processes are discontinuous, they require sorbent regeneration once saturation is reached. Therefore, optimal operation of these complex systems requires the control of many time-dependent variables. In this framework, the rapid analyses that can be carried out via dynamic mathematical models could be greatly beneficial in lowering costs at the design stage.

In this study, a model of a sorption-enhanced methanation system was developed and validated with experimental data. The proposed model allows the dynamic behavior of the process to be captured during production and sorbent regeneration. Insights into the process show the behavior of the sorbent bed while retaining water, demonstrating how the bed saturation occurs and how to effectively exploit it.

The model potential was assessed by analyzing the alternating operation of two reactors for continuous production. The importance of bed drying was underlined to ensure the enhancement of the reaction and maximize the conversion.

1. Introduction

The handling of the temporal variability of most renewable sources is one of the major challenges of the near future. Green hydrogen together with synthetic methane, ammonia, methanol [1], dimethyl ether [2], and other renewable fuels can be regarded as candidates for addressing this challenge [3]. These fuels allow the decoupling of the asynchronous energy generation and demand by storing the electrical energy produced in the chemical bonds. In these terms, it is possible to obtain a type of energy storage which can be maintained for the long term.

Among the above-mentioned fuels, methane is an attractive solution. In fact, apart from the long-term storage, it can use the existing natural gas infrastructure as well as all the technologies available today for the exploitation of natural gas. If synthesized from green hydrogen and captured carbon dioxide, net climate-altering emissions are low because the carbon dioxide that will be emitted by burning the methane, has first been sequestrated from other processes or removed directly from the atmosphere. In comparison to hydrogen, the main advantage is the energy density, which is three times higher at moderate storage pressures (around 200 bar). In addition, methane offers easier and safer handling.

The production process involves the reaction of 1 mol of carbon dioxide and 4 mol of hydrogen forming 1 mol of methane and 2 mol of water. This is commonly performed in fixed-bed catalytic reactors [4]. During the reaction, a volume reduction takes place since the number of moles of the products is smaller than that of the reactants. Therefore, due to Le Chatelier's principle, increasing pressure shifts the chemical equilibrium towards the products. In addition, the reaction is strongly exothermic and, therefore, favored at low temperatures. However, the reaction temperature cannot be too low as this would result in chemical kinetics limitations.

To make use of the existing technologies, it is important to maintain the synthesized methane quality at the same level as the fossil natural

* Corresponding author.
E-mail address: andrea.barbaresi@unipr.it (A. Barbaresi).

https://doi.org/10.1016/j.renene.2024.121020
Received 14 February 2024; Received in revised form 20 June 2024; Accepted 15 July 2024
Available online 20 July 2024
0960-1481/

gas. In fact, natural gas boilers, engines, or turbines, have been designed over the years based on the specific energy characteristics of natural gas. Therefore, the synthesis of a green gas for its replacement needs to have a calorific value comparable to or higher than that of the fossil gas, which requires the highest possible conversion rates of the reactants. To ensure this, there are some approaches in the literature that aim to obtain very high conversion rates, such as Ru-catalyzed methanation operating at lower temperatures [5] or staged reactor systems with intermediate water removal [6]. In this paper, sorption-enhanced methanation is considered. Sorption-enhanced processes are able to achieve very high conversion rates since they minimize the flow of unconverted reactants at the reactor outlet [7]. By selectively removing the water from the reaction environment, the chemical equilibrium shifts to the products side. This in-situ water separation has been reported as an approach for the enhancement of several types of equilibrium-limited reactions [8]. However, this technique requires regeneration of the bed when it reaches saturation, releasing the water by desorption. This can be performed mechanically by lowering the pressure, thermally by supplying heat, or both. When both are considered, it is referred to as a temperature and pressure swing adsorption process [9].

For the optimal operation of these discontinuous processes, the use of mathematical models capable of simulating all the phases involved can be greatly helpful. Furthermore, an a priori understanding of how such plants should be operated can drastically reduce unnecessary costs caused by failures or improper operation. Therefore, for the purpose of proper dimensioning, operation and control, the use of adequate mathematical models is essential.

Within the framework of novel approaches for the production of synthetic energy carriers, real-time simulation tools are lacking. However, such tools allow the implementation of complex and robust strategies for efficient and safe plant operation and control.

The aim of this work is to explore the possibilities of real-time simulations and to implement tools for capturing the main characteristics of sorption-enhanced synthetic methane production. Therefore, the suitable detail and discretization level needed to gather all necessary information was identified and implemented while keeping real-time simulation capability. The developed mathematical model is capable of capturing the main features of this technology by reproducing the entire production system on the basis of the behavior of all the involved components. The capability to describe all process phases can be used in real plants to define control logics and operating schedules.

2. Framework and article contribution

The sorption-enhanced intensification practice was applied to catalytic methane production initially by Borgschulte et al., in 2013 [10], and was then further developed.

In particular, energy and thermodynamic considerations were conducted by Walspurger et al. in Ref. [11] and by Massa et al. in Ref. [12] with a focus on the problem of carbon deposition. The best options for the catalytic/adsorptive material have been investigated in numerous studies. Specifically, the effect of the pore size of two different materials was investigated by Borgschulte et al. [13], different catalyst contents in the bed were tested by Wei et al. in Ref. [14], while different sorbents were analyzed by Agirre et al. in Ref. [15] and by Coppola et al. in Ref. [16]. Furthermore, a bifunctional material was tested for long-term stability by Delmelle et al. [17] and the optimal operating conditions were identified by Gómez et al. in Ref. [18] for a system with a Rh catalyst and zeolite 5A adsorbent. In addition, insights regarding the mechanism of the sorption-enhanced reaction were also identified. In particular, a coupled reaction and adsorption front was highlighted by Borgschulte et al. in Ref. [19]. In all the cited studies, it has been reported that the quality of the produced syngas is significantly improved with respect to conventional methanation processes.

Regarding the conventional methanation process, numerous works addressed the problem by means of numerical simulations. The first study [20] concerning mathematical models on the subject dates back to 1974, but research has intensified particularly in the last decade.

Depending on the purpose, studies with stationary or dynamic reactor models have been proposed, with a spatial layout ranging from zero-dimensional to 3D (e.g. Rönsch et al. [21]). They could also range from single-phase to multi-phase, in the case of models that included gas interaction with the solid phase of the reactor bed and/or possible cooling fluids.

By means of modeling and simulation it was possible to assess the viability of multiple kinds of reactors (Strucks [22]). Specifically, a circulating fluidized bed reactor was investigated numerically by Sun et al. in Ref. [23] while small-scale reactors were simulated by Moioli et al. in Ref. [24]. The use of molten salts for thermal regulation was studied in Ref. [25], and the best control strategies were defined for fixed-bed reactors by Kao et al. [26] and by Bremer et al. [27]. The off-design of this process was analyzed by Giglio et al. [28] for an intermittent operation and by Güttel [29] with respect to step responses.

Sorption-enhanced processes have also been explored through numerical approaches. Works concerning the production of methanol [30, 31] and dimethyl ether [32] are mentioned when focusing on the synthesis of energy carriers.

Few studies regard the mathematical modeling of the sorption-enhanced methanation process. Scala et al. [12] analyzed the carbon deposition and the outlet gas quality by means of thermodynamic studies using the Gibbs free energy minimization technique. The same technique was utilized in a three-reactor cascade configuration simulated in the ASPEN plus environment in Ref. [11]. The reactants breakthrough times (denoting the end of the reaction phase) were assessed in Ref. [33] by means of a two-dimensional model. Lastly, Kiefer et al. [34] used a theoretical model to support the experimental data in describing the water adsorption mechanism and the sorption-enhanced methane production.

While [11,12] consent only evaluation at equilibrium, kinetics is taken into consideration in Ref. [34] for reactor design purposes and in Ref. [33] for process scheduling. No papers dealt with the dynamic simulation of the reactor integrated into a plant with the aim of plant control and operation.

Therefore, with respect to what has been summarized from the literature, this paper describes a mathematical model which allows the simulation of the dynamic aspects of a discontinuous process such as sorption-enhanced methanation.

Specifically, the novelty of this work is that it provides a mathematical model validated on experimental data and capable of.

- investigating the dynamic behavior of sorption-enhanced methanation and, therefore, assessing the potential use of sorption-enhanced methane synthesis systems in real dynamic applications;
- capturing the main features of sorption-enhanced methanation technology, including adsorption and desorption, with good accuracy;
- taking into consideration the system in which the methanation reactor is embedded and the interaction between the various components.

To this end, all components were modeled considering the chemical and physical principles that dominate the phenomena involved. As for the reactor itself, mass and energy balances were combined with computations concerning the chemical kinetics of the reactions involved and the dynamics of adsorption and desorption.

The steps taken to achieve the intended purpose will be described throughout this paper. A detailed description of the set of model equations will be provided in section 3.1. In section 4.1, the results of the model validation on the experimental data will be illustrated by means of root mean square errors. Then, some features of the process, which can be captured by the model but are difficult to measure experimentally, will be given in section 4.2. In addition, as a demonstration of the capabilities of the model, a possible application with two reactors in alternating operation will be shown in section 4.3.

3. Materials and methods

3.1. Reactor model development

A dynamic mathematical model for a heterogeneous chemical reactor was developed. The model is based on mass and energy balance equations. In addition, chemical kinetics and adsorption dynamics were also considered.

The chemical reactor was discretized into control volumes arranged in series along the reactor axis as schematized in Fig. 1. Each control volume was characterized by a lumped parameter approach whereby the properties of the gases present were considered homogeneous. In this configuration, it is assumed that phenomena of axial diffusion are negligible since, based on estimations on thermal and species Péclet numbers, diffusive transport is in both cases negligible compared to advective transport. This assumption is commonly made in flow reactor modeling [35].

3.1.1. Chemical kinetics

The chemical reactions considered in the model are carbon dioxide methanation (R1), carbon monoxide methanation (R2), and reverse water gas shift (R3) [36].

$$CO_2 + 4H_2 \leftrightharpoons CH_4 + 2H_2O, \qquad \Delta H_r^0 = -165\ \frac{kJ}{mol} \tag{R1}$$

$$CO + 3H_2 \leftrightharpoons CH_4 + H_2O, \qquad \Delta H_r^0 = -206\ \frac{kJ}{mol} \tag{R2}$$

$$CO_2 + H_2 \leftrightharpoons CO + H_2O, \qquad \Delta H_r^0 = +41\ \frac{kJ}{mol} \tag{R3}$$

The reaction rates were computed as follows on the basis of the temperature, the thermodynamic equilibrium and the partial pressure of the species involved as proposed by Xu and Froment [37] for heterogeneous catalysis:

$$r_{R1} = \chi_{R1}\xi_{R1}\frac{k_{R1}}{p_{H_2}^{3.5}}\left(p_{CH_4}p_{H_2O}^2 - \frac{p_{H_2}^4 p_{CO_2}}{K_{R1}}\right) \Big/ (DEN)^2 \tag{1}$$

$$r_{R2} = \chi_{R2}\xi_{R2}\frac{k_{R2}}{p_{H_2}^{2.5}}\left(p_{CH_4}p_{H_2O} - \frac{p_{H_2}^3 p_{CO}}{K_{R2}}\right) \Big/ (DEN)^2 \tag{2}$$

$$r_{R3} = \chi_{R3}\xi_{R3}\frac{k_{R3}}{p_{H_2}}\left(p_{CO}p_{H_2O} - \frac{p_{H_2} p_{CO_2}}{K_{R3}}\right) \Big/ (DEN)^2 \tag{3}$$

$$DEN = 1 + A_{CO}p_{CO} + A_{H_2}p_{H_2} + A_{CH_4}p_{CH_4} + A_{H_2O}p_{H_2O}/p_{H_2} \tag{4}$$

where the terms k (reaction rate coefficients) and K (reaction equilibrium constants) are functions of temperature of the Arrhenius type, while the terms A are the adsorption constants. Each reaction rate Eqs. (1)–(3) presents two multiplicative factors, namely χ and ξ. The former is used, as will be explained in the following, to take into account the reactor model discretization, while the latter considers the utilization of a different catalyst with respect to the one used in the literature.

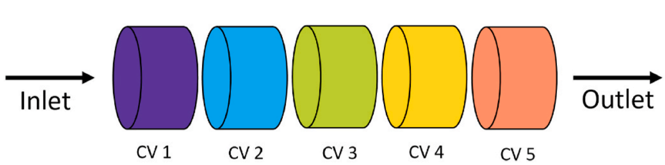

Fig. 1. Schematic representation of the reactor model discretization.

3.1.2. Adsorption dynamics

The adsorption dynamics was modeled using the approach proposed in Ref. [38] based on the solid linear driving force model:

$$\frac{\partial q_{H_2O}}{\partial t} = \frac{q_{H_2O}^* - q_{H_2O}}{\varrho_{H_2O}} \tag{5}$$

The rate of adsorption depends on the deviation of the adsorbed phase concentration q_{H_2O}, with respect to its value at equilibrium $q_{H_2O}^*$ through the coefficient ϱ_{H_2O}.

The equilibrium adsorbed phase concentration was estimated by the correlation of Dubinin-Astakhov [39]:

$$\frac{q_{H_2O}^*}{q_{m,H_2O}} = e^{-\left(\frac{RT}{M_{H_2O}E}\, ln\frac{p_{sat}}{p_{H_2O}}\right)^n} \tag{6}$$

where the water partial pressure p_{H_2O}, the saturation pressure p_{sat} at the temperature T, and the characteristic energy of the adsorbent E, are considered.

The rate of adsorption was then included in a sink term of mass flow rate and estimated by the following equation that, for the case of water only, was added to the mass balance equation:

$$\dot{m}_{ads} = (1-\varepsilon)(Sl)\rho_s\frac{\partial q_{H_2O}}{\partial t}M_{H_2O} \tag{7}$$

In Eq. (7), ε represents the void fraction of the fixed bed, S its cross section, l the thickness of the control volume in which the equation is computed and ρ_s the density of the bed solid phase.

The sorbent used, zeolite 13X, adsorbs a small amount of carbon dioxide as well. In this study, however, this effect was not considered because the amounts of adsorbed carbon dioxide compared to those of water are significantly lower [16].

The heat emitted by adsorption was evaluated through Eq. (9) and was considered to be released on the solid surface (subscript "surf") and then exchanged with the gas flow through a convective term, Eq. (10). Therefore, the temperature of the solid phase was computed with Eq. (8) as follows:

$$\frac{\partial T_{surf}}{\partial t} = \frac{\dot{Q}_{ads} + \dot{Q}_{conv}}{Sl(1-\varepsilon)(\rho_s c_s)} \tag{8}$$

$$\dot{Q}_{ads} = \rho_s Sl(1-\varepsilon)\left(-\Delta H_{ads,H_2O}\right)\frac{\partial q_{H_2O}}{\partial t} \tag{9}$$

$$\dot{Q}_{conv} = a_p(1-\varepsilon)(Sl)h_{gs}\left(T - T_{surf}\right) \tag{10}$$

3.1.3. Mass and energy balances

The mass conservation equation was written as follows:

$$(Sl)\varepsilon\frac{\partial \rho_\alpha}{\partial t} = \left(Q_{v,in}\rho_{\alpha,in} - Q_v\rho_\alpha\right) + \dot{m}_{\alpha,gen} \tag{11}$$

where the volumetric flow rate Q_v is computed taking into account the bed porosity as follows:

$$Q_v = S\varepsilon u \tag{12}$$

The velocity of the gas flow through each control volume, was computed on the basis of the pressure drop occurring from that control volume and the following one (the control volumes were arranged in series). The pressure computation in an outlet volume downstream of the reactor allowed the computation of the flow velocity in the last control volume as well. The subscript "in" refers to the control volume inlet, which corresponds to the exit of the preceding control volume. The pressure drop was computed by means of the Ergun correlation [40]:

$$f = -\frac{\Delta P \psi d_p}{l(\rho u^2)} = \frac{1-\varepsilon}{\varepsilon^3}\left(1.75 + 150\frac{1-\varepsilon}{\mathrm{Re}'}\right) \tag{13}$$

where the modified Reynolds number is defined as:

$$\mathrm{Re}' = \frac{\psi d_p \rho u}{\mu} \tag{14}$$

Eq. (11) applies to each gaseous species α. The aim is to compute the density inside the control volume. The source term $\dot{m}_{\alpha,\mathrm{gen}}$ denotes the mass flow rate generated (or removed, if negative) by means of the chemical reactions and is computed according to the following equation:

$$\dot{m}_{\alpha,\mathrm{gen}} = (Sl)\rho_{\mathrm{cat}}(1-\varepsilon)M_\alpha \sum_j \nu_{\alpha j} r_j \tag{15}$$

When considering the water, the mass balance equation includes the sink term of the adsorbed mass flow rate. In that case, the temporal density change is computed with the following equation:

$$\frac{\partial \rho_{H_2O}}{\partial t} = \frac{1}{S\varepsilon l}\left[Q_{\mathrm{v,in}}\rho_{H_2O,\mathrm{in}} - Q_{\mathrm{v}}\rho_{H_2O} + \dot{m}_{H_2O,\mathrm{gen}} - \dot{m}_{\mathrm{ads}}\right] \tag{16}$$

The energy contributions to an entire control volume were written as follows:

$$(\rho c)_{\mathrm{eff}}(Sl)\frac{\partial T}{\partial t} = \sum_\alpha \left(\dot{m}_{\alpha,\mathrm{in}} h_{\alpha,\mathrm{in}} - \dot{m}_\alpha h_\alpha\right) + \dot{Q}_{\mathrm{gen}} - \dot{Q}_{\mathrm{r}} + \dot{Q}_{\mathrm{conv}} \tag{17}$$

where the subscript “eff” denotes the effective thermal capacity of both the solid bed and the gaseous flow while $\dot{Q}_{\mathrm{gen}}$ accounts for the heat released by the exothermal reactions.

For each control volume, the radial heat flow $\dot{Q}_{\mathrm{r}}$ exchanged with the outside environment (the thermal oil flowing in the reactor jacket for thermal regulation purposes) was estimated assuming a constant global heat transfer coefficient.

Nevertheless, a higher constant value was used for the regeneration phase with respect to the production phase. This was done considering that the heat transfer coefficient of the mixture in the regeneration phase (which has high concentrations of hydrogen and water steam) is higher than that of the mixture present during the production phase.

$$\dot{Q}_{\mathrm{r}} = (\pi D l)\, U\,(T - T_{oil}) \tag{18}$$

The convective term $\dot{Q}_{\mathrm{conv}}$ indicates, as mentioned previously, the heat exchanged from the gaseous flow and the solid surface heated up by the heat of adsorption.

3.2. Models of system components

In addition to the reactor, other components of the system were modeled. These components can be divided into the following groups: tanks, pipes, valves, and control signals.

The tanks modeling follows the mass dynamics in volumes. The gas pressure variation in the tanks was expressed on the basis of the mass conservation equation and the perfect gas law as reported in Ref. [41]:

$$p = \frac{M(t)}{V}\sum_\alpha w_\alpha R_\alpha T \tag{19}$$

$$m(t) = \int_0^t \sum_\alpha \sum_l w_{\alpha,l}\dot{m}_l d\tau \tag{20}$$

where α refers to the gas species and l to the inlet/outlet flows.

The pipe model relates the pressure at the endpoints and the mass flow rate passing through the pipe by means of the following:

$$p_{\mathrm{in}} - p_{\mathrm{out}} = \frac{\dot{m}^2}{\rho A_{\mathrm{cs}}}\left(\frac{fL}{D_{\mathrm{in}}} + Z\right) \tag{21}$$

The valve model is based on the physics of flows in restrictions as proposed in Ref. [41]:

$$\dot{m} = C_v Y \sqrt{\rho_{in}\Delta p} \tag{22}$$

where the flow compressibility is taken into account by valve expansion factor Y that depends on upstream and downstream pressure.

The valve openings were controlled in order to regulate the pressure of the reactor and the mass flow rates of the gases by means of PID controllers.

3.3. Experimental setup for the model validation

In order to validate the model, the experimental apparatus previously utilized in Ref. [7] was employed. The required periodic adsorption and desorption resulted in a temperature and pressure swing adsorption procedure. Specifically, during the regeneration phase, with respect to the thermodynamic conditions set during the reaction phase, the temperature had to be increased, while the pressure had to be decreased.

The reactor consisted of four ducts containing a catalyst bed with a diameter of 32 mm and a length of 1 m (Fig. 2). The fixed bed was composed of a bi-functional material with both catalytic and adsorptive properties. Specifically, zeolite 13X pellets (ranging from 1.6 mm to 2.6 mm in diameter) were used as adsorbents and were impregnated with the nickel catalyst (5 % on a mass basis). In the initial part of the reactor, zeolite pellets were mixed with aluminum beads in order to avoid excessive thermal loads, as reported by Kiefer et al. in Ref. [34].

Source tanks of methane, hydrogen, and carbon dioxide were used upstream of the reactor. The first was used, when needed, to pressurize the system while the other two were used to prepare the reactant mixture for the reaction. Due to the inaccuracy of the hydrogen mass flow controller at the low flow rates tested, the hydrogen mass flow rate could not be set to the stoichiometric value. However, it was possible to compute it a posteriori, and the value of the amount that actually entered the reactor during the experimental campaign was included in the model. Therefore, only the carbon dioxide flux will be referred to in the following. In addition, a flow of pure hydrogen was utilized to purge the bed during the regeneration phase.

The inlet gaseous flux was regulated by means of mass flow controllers (MFC, Bronkhorst EL-FLOW Prestige). The temperature at the inlet and outlet sectors of each reactor was measured by means of thermocouples (Type K class A, 1.5 mm jacket diameter). A thermoregulator (Julabo HT60 thermostat) was used to control the reactor temperature. The regulation was carried out by means of thermal oil flowing in the external jacket of each duct. The temperature was controlled and kept at a value of 300 °C during the reaction phase, and at 340 °C during the regeneration phase. The absolute pressure was raised to 10 bar for the reaction phase and lowered to 1.5 bar for the regeneration phase. A mass spectrometer (Hiden Analytical RGA) was used for gas analysis at the reactor exit.

The model used for the validation was tailored to the experimental test rig characteristics. In addition to the reactor, other elements such as gas source tanks, connecting pipes, valves, and control signals were also modeled (see section 3.2).

3.4. Model implementation

The equations were written in a MATLAB®/Simulink® environment and solved by the solver ode15s, suitable for stiff problems, with a variable time step. The reactor model was axially discretized in control volumes, each with homogenous gas properties.

A sensitivity analysis was conducted with reference to the number of control volumes in which the component was discretized. In more detail, 5, 10, 15 and 20 control volumes were considered with different inlet reactant mole fluxes. To quantify the reactant flux, reference was made to the mole flux of CO_2 at the reactor inlet.

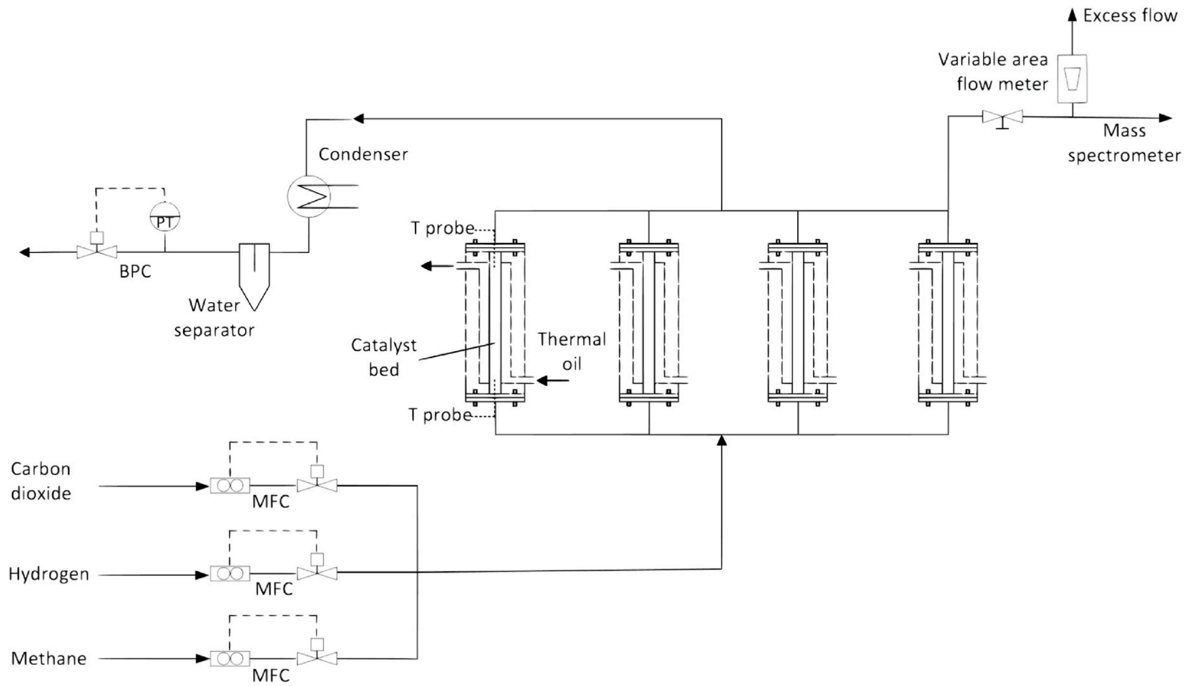

Fig. 2. Schematic representation of the sorption-enhanced methanation experimental setup used for the model validation.

Table 1
Influence of axial discretization on the computational time and the error with respect to the measurements.

Inlet CO_2 mole flux [mol m^{-2} s^{-1}]	Number of control volumes	RMS error [−]	Relative computational time
0.1	5	0.0067	1.1691
	10	0.0069	2.4246
	15	0.0069	5.2057
	20	0.0067	8.9033
0.2	5	0.0045	1.0157
	10	0.0064	2.0917
	15	0.0055	3.6779
	20	0.0060	5.7297
0.3	5	0.0154	1.0679
	10	0.0051	2.1349
	15	0.0075	3.3476
	20	0.0149	4.6488
0.4	5	0.0391	1.0000
	10	0.0076	2.1579
	15	0.0156	3.1753
	20	0.0272	5.2205

To this end, Table 1 compares the error committed and the computational time required in the different cases. The computational time was related to the smallest recorded simulation time, the one concerning a flow of 0.4 mol m^{-2} s^{-1} with 5 control volumes. The root mean square (RMS) error of the outlet methane mole fraction with respect to the experimental measurement was evaluated for the period when the production reached steady state.

As can be seen in Table 1, on the one hand, the RMS error is very small (below 0.04) for all cases and can be considered almost negligible. On the other hand, the relative computational time increased exponentially with the discretization refinement. Therefore, the analysis underlined that, for the purposes of this work and under the conditions tested, a high number of control volumes was not justified. As a result of this sensitivity analysis, the number of control volumes in which the reactor model was discretized was set to 5.

The rectifications of the chemical kinetics anticipated during the presentation of the model (section 3.1.1) will be examined below. It should be recalled that, following the model architecture, the gas concentration computed in the last control volume was regarded as the concentration leaving the reactor. The species concentration was representative of each control volume, and it is homogeneously distributed within the entire volume (lumped parameter approach).

Therefore, because of the spatial discretization of the reactor model, each reaction rate in Eqs. (1)–(3) was multiplied by the factor indicated with χ. This multiplicative factor was set inversely proportional to the control volume size (the smaller the size, the better the approximation). Reference is made to the case with the most refined discretization among those tested, in that case χ was set to 1.

The term ξ is used to adjust to the chemical kinetics in relation to the experimental catalytic system compared with the correlations used. That value is constant and was calibrated based on the experimental data considering the most refined discretization with $\chi = 1$.

In this field, the use of multiplicative factors has been suggested also by Kang and Lee [42] and for the sorption enhanced dimethyl ether synthesis by Guffanti et al. [43].

3.5. *Model application*

The model was used to test the behavior of a plant with two reactors in alternating operation. A configuration of this type is required to obtain continuous methane production. In fact, because of the need to regenerate the water-saturated bed, a single reactor is not sufficient to provide continuous methane synthesis.

A schematic diagram of the setup is provided in Fig. 3. The two methanation reactors were arranged in parallel, each of them was preceded by a volume for the mixing of the inlet gaseous flows. The gases flowing to the reactors originated from the tanks placed upstream. The flow was regulated by the valves and the mass flow controllers and then directed into one of the two reactors according to an appropriate scheduled logic.

It was imposed that the production phase of the second reactor started as soon as the one in the first reactor finished. Specifically, during the production phase of one reactor, the bed regeneration and pressurization phases took place in the other reactor and vice versa. A temperature of 300 °C and an absolute pressure of 10 bar were set during the adsorption phase, while during the regeneration phase the

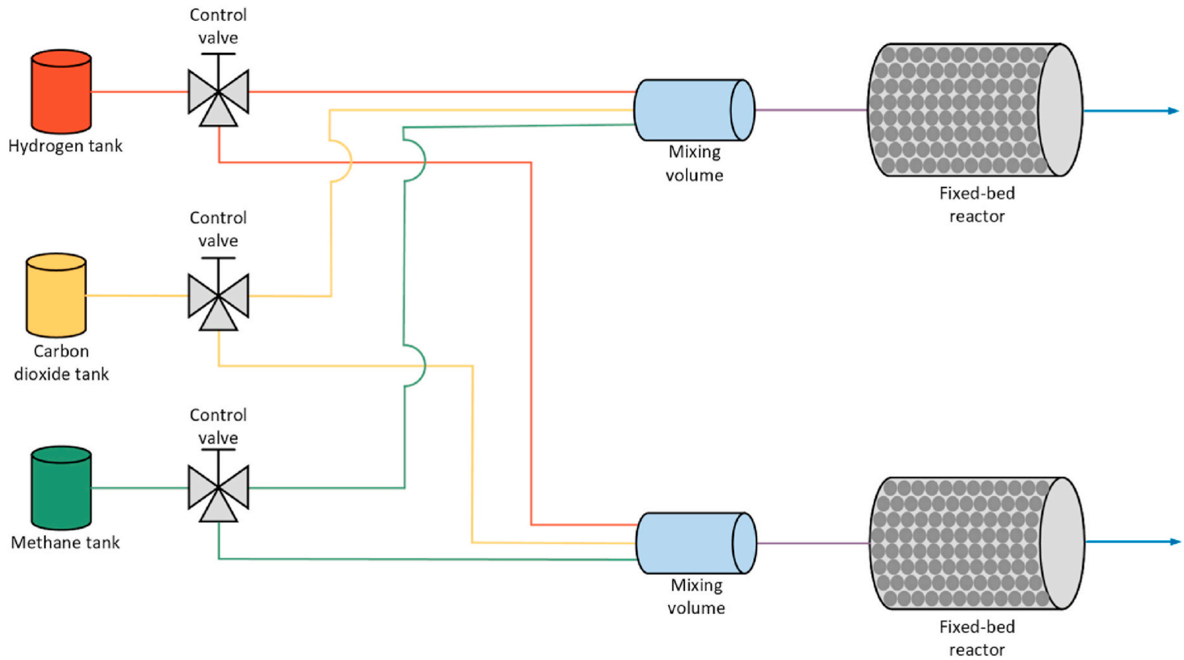

Fig. 3. Schematic representation of the modeled setup in the configuration with two parallel reactors.

temperature was increased to 340 °C and the pressure was decreased to a value of 1.5 bar.

Regarding the reactant mixture, a CO_2 mole flux of 0.2 mol m^{-2} s^{-1} and a H_2 mole flux of 0.8 mol m^{-2} s^{-1} were used for the analysis. The system was tested for 24 h of cyclic operation in which 16 production phases took place. The behavior of the system was studied by varying the hydrogen flow used for the bed regeneration.

4. Results and discussion

4.1. Model validation

In order to validate the model, experimental data from the measurements, illustrated in section 3.3, were used as the input values for the model with specific reference to the inlet gas flow rates (and, consequently, the reactant ratio), set pressures, and temperatures.

Fig. 4 shows the simulated and measured outlet mole fraction of the main gases involved in the process during both the production and regeneration phase. In this case, during the production phase the carbon

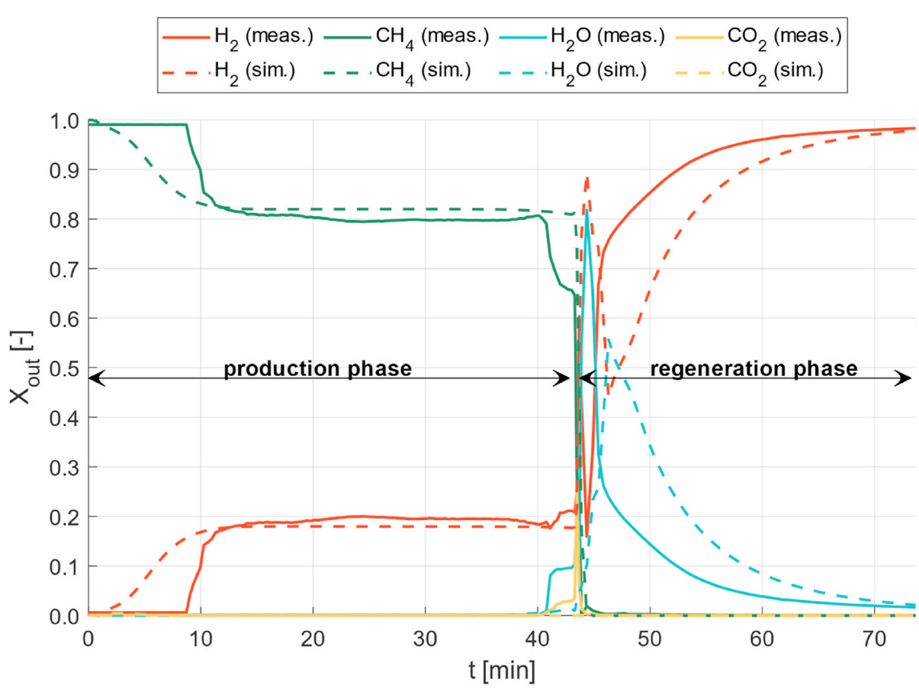

Fig. 4. Comparison of measured (meas.) and simulated (sim.) reactor outlet mole fractions (inlet CO_2 flux of 0.2 mol m^{-2} $s^{-1}$and production phase pressure of 10 bar).

dioxide flux was set at 0.2 mol m^{-2} s^{-1}, while the pressure was raised to 10 bar. The solid lines depict the measured data while the dashed lines show the model results.

From a qualitative point of view, the model was able to capture the principal trends of the process dynamics during both of the aforementioned phases.

The differences visible in the first 10 min (beginning of the production phase), between the simulated and measured outlet methane mole fraction, can be explained considering the lumped parameter approach used in the model. Indeed, in each control volume, the properties of the gas mixture were considered homogeneous. This resulted in the curve widening in the time trends. Furthermore, it is also important to underline that the simulated value referred to the last control volume and not to an element downstream of the reactor.

Between t = 10 min and t = 40 min the modeled values and the measurements are in very good agreement. At t = 43 min the production phase ended, and the regeneration phase started. Thereafter, there is some discrepancy between the simulation and the experiment. The reasons are similar to those explained for the initial production phase. However, the discrepancy decreases as the regeneration phase proceeds and becomes more stable.

Despite these limitations, the species reaction trends at the outlet were estimated with reasonable accuracy. During the steady operation phases, the accuracy is remarkably high. Furthermore, it is also noticeable that the model is able to grasp the fact that, in reactors of this type, an almost complete conversion of carbon dioxide can be achieved. This is in good agreement with the literature analyzed in section 2.

The root mean square error between the measurement and simulation results was simulated on the outlet mole fractions for four values of the inlet reactant mole flux. The results are shown in Fig. 5.

The error on the mole fraction is always below 0.02 for all the cases analyzed. It is noticeable that the absolute errors for hydrogen and methane are larger than those related to carbon dioxide and water.

The simulated temperature trends in the first (CV1) and last (CV5) control volume of the reactor model were compared with those measured in the inlet and outlet sectors and are shown in Fig. 6. The correctness of the reactor temperature estimation is extremely significant because both adsorption capacity and reaction rates depend on it, both decrease as the temperature increases. The average error computed with respect to the measured data is smaller than 5 K.

The temperature increase in CV1 (at t = 4 min) shows the start of the adsorption and reaction in this area, leading to a temperature increase to 328 °C. These tendencies are evident both in the experimental data and in the results of the simulation. During this time, CV5 stayed at 300 °C

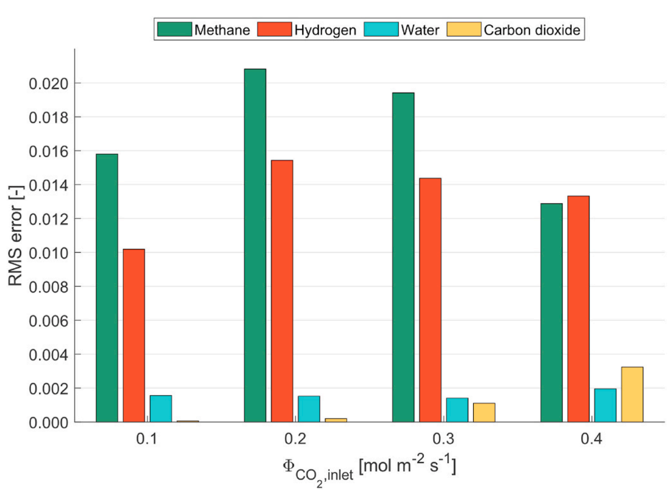

Fig. 5. *Model RMS error of the outlet mole fraction of the four main involved gaseous species with respect to the measured data for four cases with increasing inlet flux.*

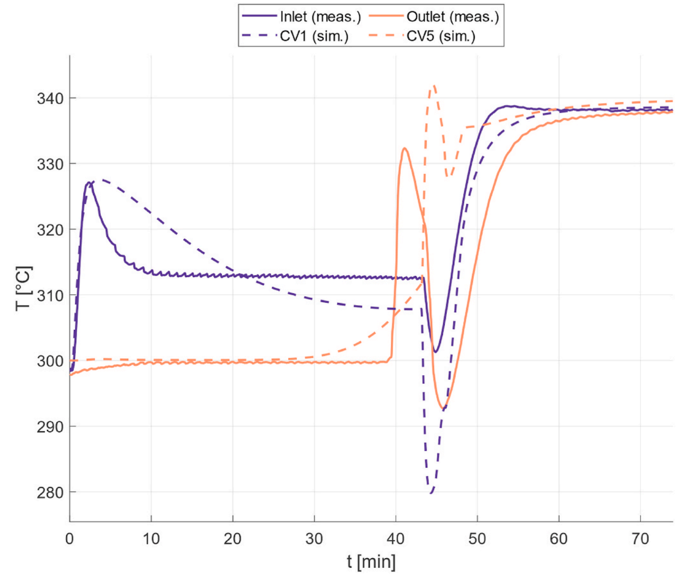

Fig. 6. Comparison of the temperature measured (meas.) at the reactor inlet and outlet and simulated (sim.) in the first (CV1) and last (CV5) control volume (inlet CO_2 flux of 0.2 mol m^{-2} $s^{-1}$and production phase pressure of 10 bar).

following the control of the thermal oil. At t = 37 min, the reaction front reached CV5. Peak temperatures here exceeded 330 °C, being slightly higher than in CV1. As discussed with respect to Fig. 4, the curve widening in the time trends is due to the spatial discretization employed. The temperature evolution inside the reactor is discussed in Fig. 7 in detail.

Following the end of production (start of regeneration) there are some discrepancies between the simulation and experiment. In the simulation, the change in temperature of the coolant was instantaneous, while the thermal oil used in the experiments had non-negligible thermal inertia.

By reaching steady conditions during regeneration, the reactor temperature homogeneously reached the control temperature of the thermal oil.

4.2. Process insights

The model was also able to simulate the progression of the temperature front that was measured in Refs. [19,34], here shown in Fig. 7. The x-axis displays the sequential number of control volumes along the

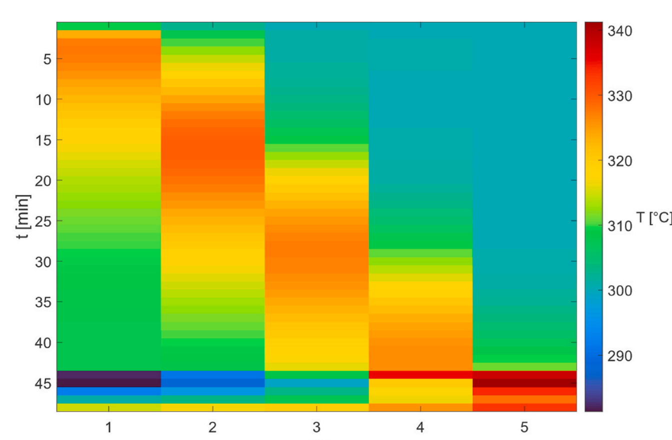

Fig. 7. Simulated temperatures in each modeled control volume during the timespan of the production phase (inlet CO_2 flux of 0.2 mol m^{-2} $s^{-1}$and pressure of 10 bar).

reactor.

The temperature front moved in time and along the reactor axis due to the coupled reaction and adsorption front, reported in the literature in Ref. [34]. The front peak reached temperature values that exceed 320 °C.

At t = 43 min regeneration started in CV1 leading to a temperature decrease, while CV5 was still at the end of the production phase reaching the temperature peak. The shift to a higher global heat transfer coefficient during the regeneration phase is the cause of the temperature drop visible after 43 min in the first two control volumes.

Fig. 8 shows the cumulative mass of water (the second reaction product besides methane) adsorbed in the bed of each control volume. During the production phase (first 43 min) the adsorption evolved, then the desorption followed. Adsorption was stronger in CV1, however, some water soon reached CV2 and then CV3. Therefore, the model highlighted the fact that some water flowed over the reaction front. A similar effect was described in Ref. [34]. It is possible to note the tendency in each control volume to reach the maximum value of sorption capacity that, in the configuration investigated, settled at approximatively 5.6 mg for 1 kg of catalyst bed. This maximum value was reached only in CV1 and CV2. To minimize the water spill-over, the production phase had to be stopped at t = 43 min, and therefore, the sorption capacity of CV3, CV4 and CV5 could not be fully exploited. In fact, only a small fraction of CV5 was used for adsorption purposes. Following the saturation of CV2, the high water loads reaching the following CVs caused an increase in the adsorption rates. Therefore, in Fig. 8, the slope of the curve rapidly changed in CV3 to CV5. This change in slope was less pronounced in CV3 and more visible in CV4 and CV5.

After t = 43 min, when the desorption started, the CV1 was the fastest to dry since it was continuously fed by a dry hydrogen flow. Downstream CVs dried at slower rates because the flow entering each CV had increasing water loads.

4.3. Application insights

The dynamic operation of a sorption-enhanced methanation system designed for a continuous methane production was investigated with the developed model. To this end, as anticipated in section 3.5, a configuration with two parallel reactors was adopted. The investigations focused on the examination of different regeneration flows, with the aim to identify the lowest flow of hydrogen leading to optimal drying. Therefore, several simulations were carried out, with the hydrogen regeneration flow as the main parameter.

The plant (see Fig. 3) was operated, alternating periodically between

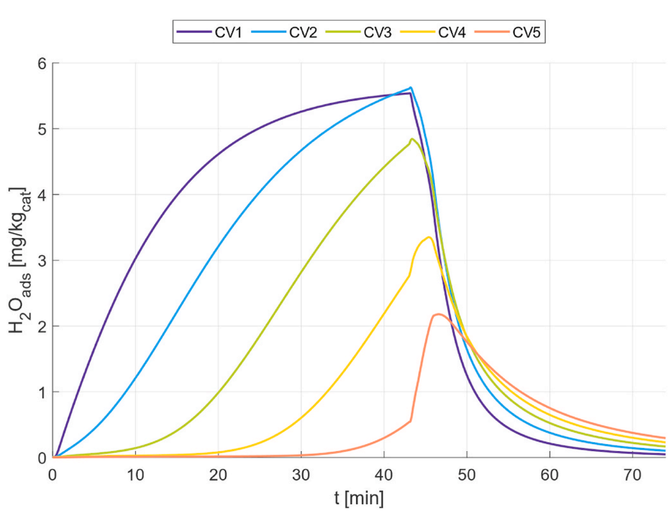

Fig. 8. Cumulative adsorbed water mass in each modeled control volume (inlet CO_2 flux of 0.2 mol m^{-2} $s^{-1}$and production phase pressure of 10 bar).

the production and regeneration phases as shown in Fig. 9. When one reactor was in the production phase, the bed regeneration (drying with a hydrogen stream) and following pressurization (using a methane stream) were carried out in the other reactor. The inlet flow of the reactants during the production phase had a stoichiometric ratio corresponding to a CO_2 flux of 0.2 mol m^{-2} s^{-1} and to a H_2 flux of 0.8 mol m^{-2} s^{-1}. The duration of the sorption-enhanced reaction for the reactors with these fluxes and geometry was considered according to Ref. [7]. Specifically, 43 min was the time needed for the production phase and 31 min for the regeneration phase as schematized in Fig. 9.

With this schedule, each reactor was able to complete 16 production cycles within 24 h. The simulations showed that the investigated system was able to produce about 250.48 kg/(m^3·day) of methane per unit volume of the reactor. Moreover, 13.42 kg/(m^3·day) of hydrogen were used for regeneration and recirculated during each cycle, while 13.36 kg/(m^3·day) of methane were used to pressurize the system (and reused at each cycle).

Fig. 10 shows the H_2O mole fraction computed at the outlet of one of the two reactors (i.e. in CV5) for different flows of hydrogen used for regeneration at three key points in the cycle: at the end of the production phase, at the end of the regeneration phase and at the end of the pressurization phase. The last one corresponds to the beginning of the forthcoming production phase.

With regard to the end of regeneration (red curve) and to the end of pressurization (yellow curve), with increasing regeneration flow, the molar fraction of water exiting the reactor decreased due to its stronger drying potential.

It is observed from the different values reached at the end of regeneration (red curve) and at the end of pressurization (yellow curve) that, under the conditions investigated, the pressurization phase plays an important role in the bed drying (resulting in wet methane during the pressurization phase).

Referring to the end of the production phase (blue curve), it is possible to observe the amount of water, generated during the production phase, reaching the outlet. The values strongly depend on the starting condition of production, which corresponds to the end of pressurization (yellow curve).

At hydrogen flows higher than 0.5 mol m^{-2} s^{-1}, the water content at the end of pressurization (yellow curve) was close to 0, while the corresponding water amount at the end of the production phase (blue curve) stabilized at values around 0.4. On the other hand, at low hydrogen regeneration flows, the slightly higher values attained at the end of the pressurization phase related to higher values at the end of production.

The amount of water left in the bed in each control volume at the beginning of a new production cycle (after regeneration and pressurization with the considered schedule) is shown in Fig. 11 together with the corresponding degree of saturation, reported as σ.

As could be expected, the amount of residual water left adsorbed in the bed differed depending on the axial location in the reactor. The first control volume (CV1) was the one that always encountered a completely dry flow, and, consequently, had the lowest amount of remaining water after regeneration and pressurization. As the position on the reactor axis moved to the exit (and the control volume number increases), the drying stream of hydrogen contained increasing amounts of remaining water. The humidity in the drying stream allowed less water to be removed from the bed but was also counterproductive as some of it was adsorbed by bed portions that were not fully saturated. Therefore, each downstream control volume was less dried and thus retained more water than the preceding one at the start of a new adsorption cycle.

It is also apparent that at low regeneration flows it was not possible to thoroughly dry the bed. In that case, the amount of residual water left adsorbed in the bed was not negligible. This results in the fact that the enhancement effect due to adsorption was reduced as illustrated in Fig. 12.

The figure shows a parametric analysis regarding the outlet mole

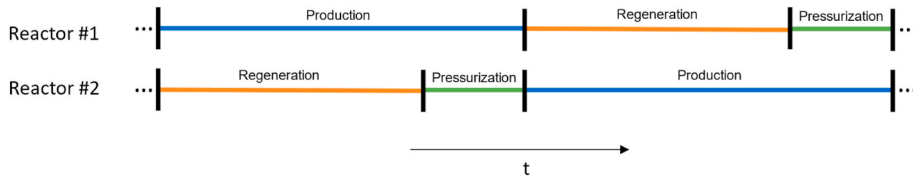

Fig. 9. Schematic representation of the alternating operation phases of the two fixed-bed reactors analyzed, according to the values in Ref. [7].

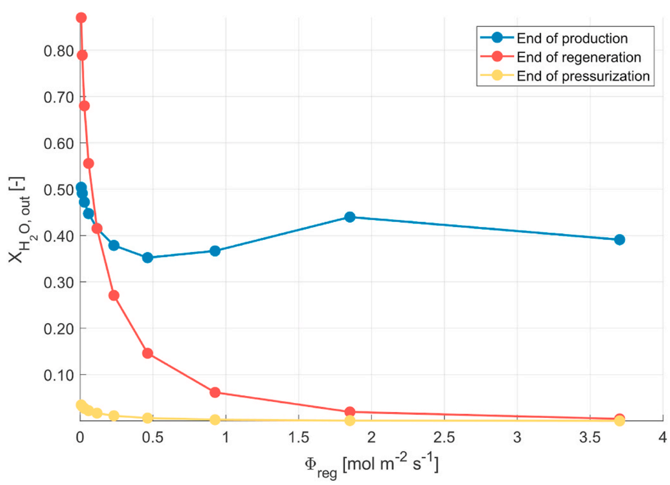

Fig. 10. Water mole fraction in the outlet gas simulated at the end of production, end of regeneration and end of pressurization for different H_2 regeneration flow values.

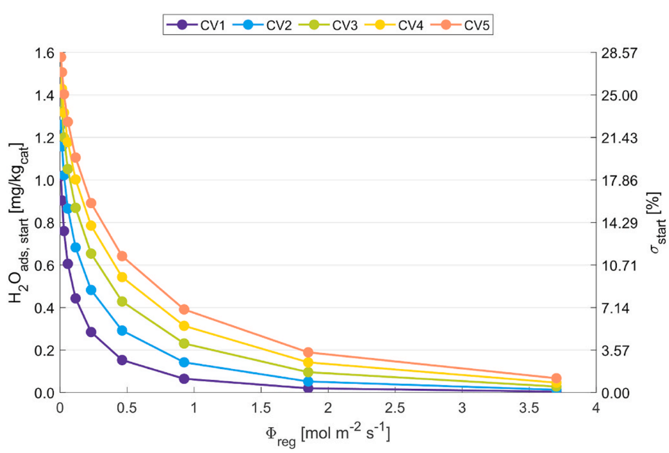

Fig. 11. Quantity of water remaining adsorbed in the bed and the corresponding degree of saturation at the beginning of a new production cycle as a function of different hydrogen fluxes in the previous regeneration phase.

fractions of methane, hydrogen, carbon dioxide and water with an increasing degree of saturation. In particular, the degree of saturation of the first control volume (CV1) at the beginning of the production phase is displayed on the x-axis, while the outlet mole fraction values averaged over the production phase period are reported on the y-axis.

As the degree of saturation of the first control volume increases, a decreasing trend for methane (product) and an increasing trend for both hydrogen and carbon dioxide (reactants) can be seen. This denotes a lower conversion capability of the reactors since a humid bed leads to a lower adsorption rate (see Eq. (5)) and to a lower total amount of adsorbed water.

Therefore, a humid bed fails to enhance the reaction with respect to a thoroughly dried bed at the beginning of the production phase. When the value of the outlet hydrogen and carbon dioxide exceeds the allowable limits for injection of the synthesized gas into the grid, a downstream gas cleaning unit should be added to the system. This highlights the importance of bed drying to enhance the purity of the produced gas.

5. Conclusions

Methane synthesized from hydrogen and captured carbon dioxide has the potential to substitute natural gas in existing infrastructures and many available technologies. Obtaining the required high-purity synthetic methane is possible with sorption-enhanced processes. Innovative and complex technologies such as sorption-enhanced methanation requires the development of appropriate control and management strategies which can be implemented properly with real-time simulation tools.

In this context, the work at hand proposes a dynamic model capable of simulating the processes in a sorption-enhanced plant for the synthesis of methane in real-time. Indeed, with reference to the discontinuous process operation of this technology, both the production (methane synthesis) and bed regeneration phases were simulated with a mathematical model.

The model of a methanation reactor was included in a complex system with other components, making it possible to reproduce their interaction. Modeling the system as a whole, it was possible to tailor the overall model to an experimental system and validate it based on the experimental data.

During the relevant phases of the process (production and regeneration), the developed model was able to predict with reasonable accuracy (i) the concentrations of all relevant species involved in the process, (ii) the temperature front evolution along the reactor, and (iii) the bed regeneration dynamics.

In accordance with experiments in the literature, model results confirm that, with these systems, an almost complete CO_2 conversion can be reached. By computing the cumulative mass of water retained in the adsorbent bed, it was found that it is not possible to exploit the full available water storage capacity. While the upstream reactor part can reach complete saturation, the downstream part of the reactor bed can only be partly utilized due to the water spill-over to the outlet.

Furthermore, the model was applied to simulate a system capable of producing methane continuously through the use of two reactors in alternating operation. The importance of proper bed drying prior to the production phase was highlighted. The degree of saturation has been introduced as the ratio between the amount of water retained in the bed and the saturation value under the thermodynamic conditions investigated, since it was identified as the most relevant parameter.

As the degree of saturation increases, the enhancement of the reaction, due to the water removal, decreases. This, in turn, leads to a decrease in the system conversion as well. The unconverted reactants in the synthesized gas deteriorate its purity and may compromise the level of quality required for its direct injection into the natural gas network. Nevertheless, when properly operated these systems allow for the production of very high purity methane.

For future research, the proposed model can represent a useful tool for the development of optimal operation strategies as well as system layout improvements. Broader research perspectives based on this work

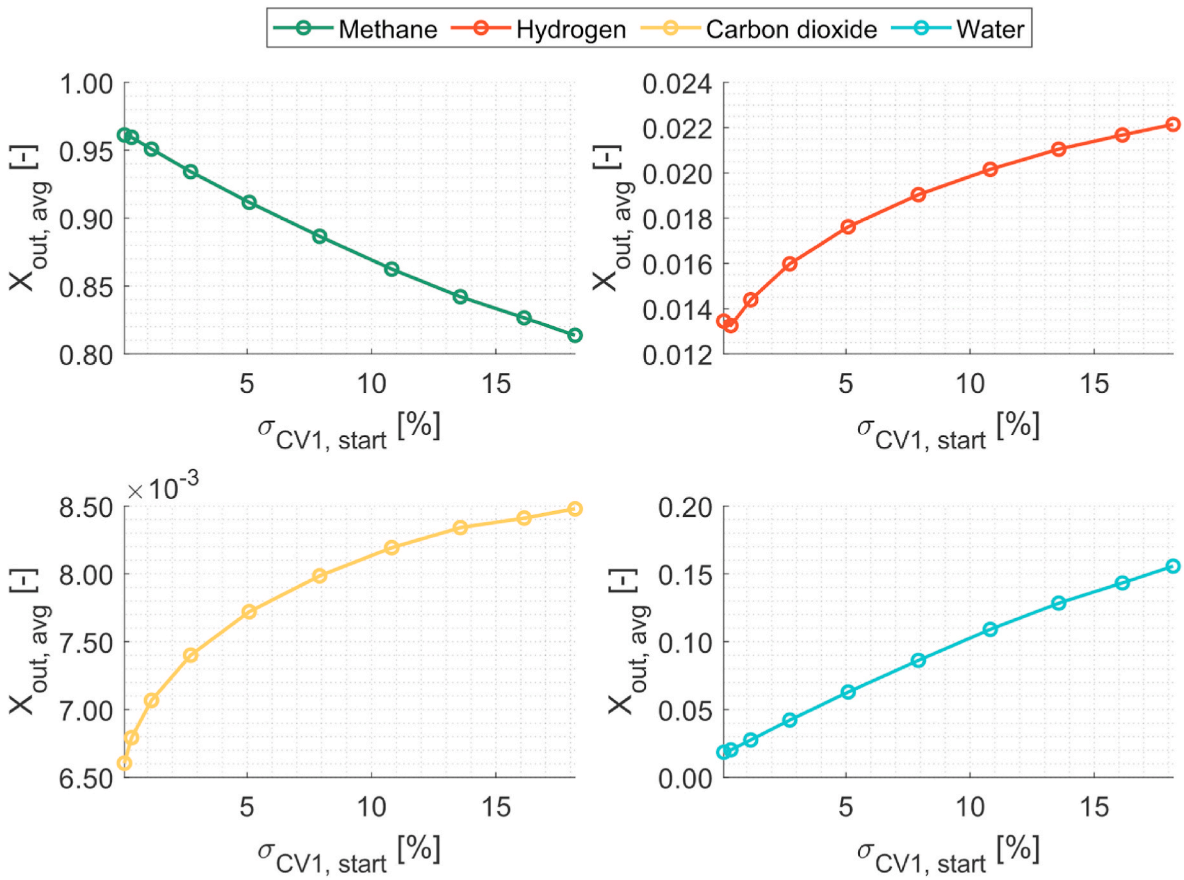

Fig. 12. Outlet mole fraction of the main species involved in the process averaged over the production phase with an increasing degree of saturation of the first control volume at the beginning of the production phase.

may involve the use of similar models in integrated energy system models to evaluate dynamic operation performance when coupling different energy vectors.

CRediT authorship contribution statement

Andrea Barbaresi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Mirko Morini:** Writing – review & editing, Supervision, Methodology. **Agostino Gambarotta:** Writing – review & editing, Supervision. **Florian Kiefer:** Writing – review & editing, Validation, Supervision, Resources. **Panayotis Dimopoulos Eggenschwiler:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgments

This work has been co-funded by a grant of European Union - PON Ricerca e Innovazione 2014–2020.

Nomenclature

A	Surface area [m^2]
a_p	Specific area of adsorbent [m^2/m^3]
A_α	Kinetics adsorption constant for species [α]
c	Specific heat [J/kg K]
CV	Control volume
C_v	Valve mass flow rate coefficient
D	Diameter [m]
d_p	Particle equivalent diameter [m]
E	Characteristic energy of the adsorbent [J/mol]
f	Friction factor
h	Specific enthalpy [J/kg]
h_{gs}	Film heat transfer coefficient between gas and solid [W/(m^2 K)]
K	Reaction equilibrium constant
k	Reaction rate coefficient [kmol/(kg$_{cat}$ h bar)]
L	Length [m]

l	Control volume thickness [m]
$\dot{m}$	Mass flow rate [kg/s]
m	Mass [kg]
MFC	Mass Flow Controller
M_{α}	Molar mass of the species α [m/s]
n	Dubinin-Astakhov exponent
p	Pressure [bar]
PID	Proportional integral derivative
p_{α}	Partial pressure of the species α [bar]
$\dot{Q}$	Thermal power [W]
q^*	Equilibrium adsorbed phase concentration [mol/kg]
q_{H2O}	Water adsorbed phase concentration [mol/kg]
q_m	Maximum adsorbed phase concentration [mol/kg]
Q_v	Volumetric flow rate [m^3/s]
r	Reaction rate [kmol/(kg_{cat} h)]
R	Universal gas constant [J/(K mol)]
RMS	Root mean square
S	Reactor cross section n area [m^2]
T	Temperature [K]
t	Time [s]
U	Global heat transfer coefficient [W/(m^2 K)]
u	Velocity [m/s]
V	Volume [m^3]
w	Gas fraction
X	Mole fraction
Y	Valve expansion factor
Z	Concentrated pressure drop factor
ΔH_r^0	Standard reaction enthalpy[kJ/mol]
ΔH_{ads}	Heat of adsorption [J/mol]

Greek symbols

α	Gaseous species
ε	Void fraction
μ	Dynamic viscosity [Pa s]
ξ	Reaction rate multiplicative factor
ρ	Density [kg/m^3]
ϱ	Mass diffusion resistance parameter [s]
σ	Degree of saturation [%]
τ	Time variable
χ	Discretization multiplicative factor
ψ	Form factor
Φ	Mole flux [mol/m^2s]

Subscripts and superscripts

ads	Adsorption
avg	Average
cat	Catalyst bed
conv	Convective
cs	Cross section
eff	Effective
g	Gas phase
gen	Generated
in	Inlet
j	Reaction index
l	Summation index
oil	Thermal oil
out	Outlet
r	Radial
reg	Regeneration
s	Solid phase
sat	Saturation
start	Beginning of the production phase
surf	Solid surface

References

[1] D. Bellotti, M. Rivarolo, L. Magistri, A comparative techno-economic and sensitivity analysis of Power-to-X processes from different energy sources, Energy Convers. Manag. 260 (May 2022) 115565, https://doi.org/10.1016/j.enconman.2022.115565.

[2] V. Dieterich, A. Buttler, A. Hanel, H. Spliethoff, S. Fendt, Power-to-liquid *via* synthesis of methanol, DME or Fischer–Tropsch-fuels: a review, Energy Environ. Sci. 13 (10) (2020) 3207–3252, https://doi.org/10.1039/D0EE01187H.

[3] M. Sterner, M. Specht, Power-to-Gas and power-to-X—the history and results of developing a new storage concept, Energies 14 (20) (Oct. 2021) 6594, https://doi.org/10.3390/en14206594.

[4] K. Ghaib, K. Nitz, F. Ben-Fares, Chemical methanation of CO_2 : a review, ChemBioEng Rev. 3 (6) (Dec. 2016) 266–275, https://doi.org/10.1002/cben.201600022.

[5] E. Moioli, A. Züttel, A model-based comparison of Ru and Ni catalysts for the Sabatier reaction, Sustain. Energy Fuels 4 (3) (2020) 1396–1408.

[6] G. Iaquaniello, S. Setini, A. Salladini, M. De Falco, CO2 valorization through direct methanation of flue gas and renewable hydrogen: a technical and economic assessment, Int. J. Hydrogen Energy 43 (36) (Sep. 2018) 17069–17081, https://doi.org/10.1016/j.ijhydene.2018.07.099.

[7] A. Barbaresi, F. Kiefer, M. Morini, P. Dimopoulos Eggenschwiler, A. Gambarotta, Partial Load and Transient Operation of Methane Synthesis Reactors Using Sorption-Enhanced Catalysis, in: Proceedings of 36th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems, ECOS 2023, 2023. https://doi.org/10.52202/069564-0086.

[8] J. Van Kampen, J. Boon, F. Van Berkel, J. Vente, M. Van Sint Annaland, Steam separation enhanced reactions: review and outlook, Chem. Eng. J. 374 (Oct. 2019) 1286–1303, https://doi.org/10.1016/j.cej.2019.06.031.

[9] B.T. Carvill, J.R. Hufton, M. Anand, S. Sircar, Sorption-enhanced reaction process, AIChE J. 42 (10) (Oct. 1996) 2765–2772, https://doi.org/10.1002/aic.690421008.

[10] A. Borgschulte, N. Gallandat, B. Probst, R. Suter, E. Callini, D. Ferri, Y. Arroyo, R. Erni, H. Geerlings, A. Züttel, Sorption enhanced CO2 methanation, Phys. Chem. Chem. Phys. 15 (24) (2013) 24, https://doi.org/10.1039/c3cp51408k.

[11] S. Walspurger, G.D. Elzinga, J.W. Dijkstra, M. Sarić, W.G. Haije, Sorption enhanced methanation for substitute natural gas production: experimental results and thermodynamic considerations, Chem. Eng. J. 242 (Apr. 2014) 379–386, https://doi.org/10.1016/j.cej.2013.12.045.

[12] F. Massa, A. Coppola, F. Scala, A thermodynamic study of sorption-enhanced CO2 methanation at low pressure, J. CO2 Util. 35 (Jan. 2020) 176–184, https://doi.org/10.1016/j.jcou.2019.09.014.

[13] A. Borgschulte, E. Callini, N. Stadie, Y. Arroyo, M.D. Rossell, R. Erni, H. Geerlings, A. Züttel, D. Ferri, Manipulating the reaction path of the CO_2 hydrogenation reaction in molecular sieves, Catal. Sci. Technol. 5 (9) (2015) 4613–4621, https://doi.org/10.1039/C5CY00528K.

[14] L. Wei, H. Azad, W. Haije, H. Grenman, W. De Jong, Pure methane from CO2 hydrogenation using a sorption enhanced process with catalyst/zeolite bifunctional materials, Appl. Catal. B Environ. 297 (Nov. 2021) 120399, https://doi.org/10.1016/j.apcatb.2021.120399.

[15] I. Agirre, E. Acha, J.F. Cambra, V.L. Barrio, Water sorption enhanced CO2 methanation process: optimization of reaction conditions and study of various sorbents, Chem. Eng. Sci. 237 (Jun. 2021) 116546, https://doi.org/10.1016/j.ces.2021.116546.

[16] A. Coppola, F. Massa, P. Salatino, F. Scala, Evaluation of two sorbents for the sorption-enhanced methanation in a dual fluidized bed system, Biomass Convers. Biorefinery 11 (1) (Feb. 2021) 111–119, https://doi.org/10.1007/s13399-020-00829-4.

[17] R. Delmelle, J. Terreni, A. Remhof, A. Heel, J. Proost, A. Borgschulte, Evolution of water diffusion in a sorption-enhanced methanation catalyst, Catalysts 8 (9) (Aug. 2018) 341, https://doi.org/10.3390/catal8090341.

[18] L. Gómez, I. Martínez, M.V. Navarro, T. García, R. Murillo, Sorption-enhanced CO and CO2 methanation (SEM) for the production of high purity methane, Chem. Eng. J. 440 (Jul. 2022) 135842, https://doi.org/10.1016/j.cej.2022.135842.

[19] A. Borgschulte, R. Delmelle, R.B. Duarte, A. Heel, P. Boillat, E. Lehmann, Water distribution in a sorption enhanced methanation reactor by time resolved neutron imaging, Phys. Chem. Chem. Phys. 18 (26) (2016) 26, https://doi.org/10.1039/C5CP07686B.

[20] P.J. Lunde, Modeling, simulation, and operation of a sabatier reactor, Ind. Eng. Chem. Process Des. Dev. 13 (3) (Jul. 1974) 226–233, https://doi.org/10.1021/i260051a007.

[21] S. Rönsch, J. Schneider, S. Matthischke, M. Schlüter, M. Götz, J. Lefebvre, P. Prabhakaranand, S. Bajohr, Review on methanation – from fundamentals to current projects, Fuel 166 (Feb. 2016) 276–296, https://doi.org/10.1016/j.fuel.2015.10.111.

[22] P. Strucks, L. Failing, S. Kaluza, A short review on Ni-catalyzed methanation of CO_2: reaction mechanism, catalyst deactivation, dynamic operation, Chem. Ing. Tech. 93 (10) (Oct. 2021) 1526–1536, https://doi.org/10.1002/cite.202100049.

[23] L. Sun, K. Luo, J. Fan, Three-dimensional simulation of the methanation process in a circulating fluidized-bed reactor, Ind. Eng. Chem. Res. 60 (45) (Nov. 2021) 16417–16429, https://doi.org/10.1021/acs.iecr.1c03107.

[24] E. Moioli, N. Gallandat, A. Züttel, Model based determination of the optimal reactor concept for Sabatier reaction in small-scale applications over Ru/Al2O3, Chem. Eng. J. 375 (Nov. 2019) 121954, https://doi.org/10.1016/j.cej.2019.121954.

[25] D. Sun, D.S.A. Simakov, Thermal management of a Sabatier reactor for CO2 conversion into CH4: simulation-based analysis, J. CO2 Util. 21 (2017) 368–382, https://doi.org/10.1016/j.jcou.2017.07.015.

[26] Y.-L. Kao, P.-H. Lee, Y.-T. Tseng, I.-L. Chien, J.D. Ward, Design, control and comparison of fixed-bed methanation reactor systems for the production of substitute natural gas, J. Taiwan Inst. Chem. Eng. 45 (5) (Sep. 2014) 2346–2357, https://doi.org/10.1016/j.jtice.2014.06.024.

[27] J. Bremer, K.H.G. Rätze, K. Sundmacher, CO_2 methanation: optimal start-up control of a fixed-bed reactor for power-to-gas applications, AIChE J. 63 (1) (Jan. 2017) 23–31, https://doi.org/10.1002/aic.15496.

[28] E. Giglio, R. Pirone, S. Bensaid, Dynamic modelling of methanation reactors during start-up and regulation in intermittent power-to-gas applications, Renew. Energy 170 (Jun. 2021) 1040–1051, https://doi.org/10.1016/j.renene.2021.01.153.

[29] R. Güttel, Study of unsteady-state operation of methanation by modeling and simulation, Chem. Eng. Technol. (Sep. 2013), https://doi.org/10.1002/ceat.201300223.

[30] Z. Dehghani, M. Bayat, M.R. Rahimpour, Sorption-enhanced methanol synthesis: dynamic modeling and optimization, J. Taiwan Inst. Chem. Eng. 45 (4) (Jul. 2014) 1490–1500, https://doi.org/10.1016/j.jtice.2013.12.001.

[31] E. Moioli, T. Schildhauer, Tailoring the reactor properties in the small-scale sorption-enhanced methanol synthesis, Chem. Ing. Tech. (2023), https://doi.org/10.1002/cite.202200200.

[32] I. Iliuta, M.C. Iliuta, F. Larachi, Sorption-enhanced dimethyl ether synthesis—multiscale reactor modeling, Chem. Eng. Sci. 66 (10) (May 2011) 2241–2251, https://doi.org/10.1016/j.ces.2011.02.047.

[33] P. Bareschino, G. Piso, F. Pepe, C. Tregambi, E. Mancusi, Numerical modelling of a sorption-enhanced methanation system, Chem. Eng. Sci. 277 (Aug. 2023) 118876, https://doi.org/10.1016/j.ces.2023.118876.

[34] F. Kiefer, M. Nikolic, A. Borgschulte, P. Dimopoulos Eggenschwiler, Sorption-enhanced methane synthesis in fixed-bed reactors, Chem. Eng. J. 449 (2022) 137872, https://doi.org/10.1016/j.cej.2022.137872.

[35] F.L. Dryer, F. M Haas, J. Santner, T.I. Farouk, M. Chaos, Interpreting chemical kinetics from complex reaction–advection–diffusion systems: modeling of flow reactors and related experiments, Prog. Energy Combust. Sci. 44 (2014) 19–39, https://doi.org/10.1016/j.pecs.2014.04.002.

[36] M. Götz, J. Lefebvre, F. Mörs, A. McDaniel Koch, F. Graf, S. Bajohr, R. Reimert, T. Kolb, Renewable Power-to-Gas: a technological and economic review, Renew. Energy 85 (2016) 1371–1390, https://doi.org/10.1016/j.renene.2015.07.066.

[37] J. Xu, G.F. Froment, Methane steam reforming, methanation and water-gas shift: I. Intrinsic kinetics, AIChE J. 35 (1) (1989) 1, https://doi.org/10.1002/aic.690350109.

[38] O.T. Qazvini, S. Fatemi, Modeling and simulation pressure–temperature swing adsorption process to remove mercaptan from humid natural gas; a commercial case study, Sep. Purif. Technol. 139 (Jan. 2015) 88–103, https://doi.org/10.1016/j.seppur.2014.09.031.

[39] M. Dubinin, V. Astakhov, Description of adsorption equilibria of vapors on zeolites over wide ranges of temperature and pressure, ACS Publications (1971), https://doi.org/10.1021/ba-1971-0102.ch044.

[40] S. Ergun, A.A. Orning, Fluid flow through randomly packed columns and fluidized beds, Ind. Eng. Chem. 41 (6) (1949) 6.

[41] A. Gambarotta, M. Manganelli, M. Morini, A model for the simulation of the gas cleaning system in a syngas-fed CHP plant, in: AIP Conference Proceedings, AIP Publishing, 2019. https://doi.org/10.1063/1.5138817.

[42] J.W. Hutchinson. ISA Handbook of Control Valves, Instrument Society of America, Pennsylvania, 1971. ISBN 0876642342.

[43] W.R. Kang, K.B. Lee, Effect of operating parameters on methanation reaction for the production of synthetic natural gas, Kor. J. Chem. Eng. 30 (7) (Jul. 2013) 1386–1394, https://doi.org/10.1007/s11814-013-0047-2.